A sub-Cohn Grassmannian Frame Packing for Cell (d=4, n=48)

Game of Sloanes catalog submission · companion technical note

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1. Summary

We report a new packing of 48 unit vectors in $\mathbb{C}^4$ with coherence $\mu = \max_{i<j} |\langle v_i, v_j\rangle| = 0.643425504760055$, improving upon the standing Cohn catalog holder for cell (4, 48) hlc, which records $\mu = 0.643427715576853$. The submission improves the holder by 2.21×10^{-6} in absolute coherence and, more importantly for the Game of Sloanes acceptance rule, by 22 units at the eighth decimal place: round-8 = 0.64342550 versus holder round-8 = 0.64342772. The submission satisfies the Game of Sloanes 8th-decimal-place acceptance rule by a comfortable margin.

The improvement was obtained on the first day of attacking this cell and is the deepest the available portfolio of optimisation paradigms could reach. The author would have preferred to deepen the packing further before filing; after a wide portfolio of paradigm families had been applied to the Cohn configuration, to the submitted configuration as anchor, and to independent random initialisations, no method available to this work produced any further descent below the value reported above.

The submission is filed because the margin is real, reproducible at the eighth decimal place across five independent verification kernels, and consistent with — though not a proof of — the working hypothesis that this basin's floor has been reached at machine-quantum precision. No claim of global optimality is made for the cell. No claim of having proven the local floor of the basin is made either.

2. The packing

The packing is provided in the accompanying file 4x48_ral.txt in the standard Game of Sloanes layout: 384 floating-point numbers, comprising 192 real components followed by 192 imaginary components for 48 complex 4-vectors. The packing was produced by the Rasputin engine — a single-threaded C++ engine operating in quad-precision (128-bit) arithmetic throughout its inner loop, on a Mac M2 laptop under 25% CPU throttling. The distinctive feature of the engine relative to standard double-precision implementations is that quad-precision arithmetic allows descent to be resolved below the threshold at which conventional double-precision engines saturate. For a cell whose holder configuration is structurally tight in the sense of admitting many simultaneously near-saturated pairs — as the Cohn holder for (4, 48) is, with 24.29% of pairs within 10^{-8} of the maximum at the holder configuration — quad-precision is the difference between stalling above the holder's coherence and being able to push to the floor of the basin in which the optimisation is conducted.

Quantity	**Value**
μ (submission)	0.643425504760055
μ (Cohn catalog holder)	0.643427715576853
Δμ (submission − holder)	-2.21×10^{-6}
Round-8 (submission)	0.64342550
Round-8 (holder)	0.64342772
Engine (this submission)	Rasputin (quad-precision)
Worst pair (stable across 5 kernels)	(1, 24)
Welch lower bound (4, 48)	0.4837794468
Levenshtein-2 bound (4, 48)	0.5877538100
Cushion below 8-dec. boundary 0.643425505	4.92×10^{-10}
Saturated-pair fraction at 10^{-8} tol.	274 / 1128 (24.29%)
Unit-norm deviation across 48 vectors	$\leq 2.21 \times 10^{-11}$
Number of floats in 4x48_ral.txt	384 = 2 · 4 · 48

3. Verification by five independent kernels

The submitted packing has been ratified byte-exact by five algorithmically distinct verification kernels, in two programming languages, using four different numerical libraries (and one path with no library at all). All five report identical coherence to within one unit in the last place of binary64 ($\approx 2 \times 10^{-16}$), which is the floor of double-precision arithmetic itself.

Kernel	**Implementation**	**μ (full precision)**	**round-8**	**pair**
K1	NumPy conj() @ T with BLAS matmul	0.6434255047600553	0.64342550	(1, 24)
K2	Pure-Python double loop, no library	0.6434255047600552	0.64342550	(1, 24)

K3	numpy.vdot per pair	0.6434255047600553	0.64342550	(1, 24)
K4	mpmath at 50 decimal digits	0.6434255047600551	0.64342550	(1, 24)
K5	Python decimal at 40 decimal digits	0.6434255047600551	0.64342550	(1, 24)

mpmath reference at 50 decimal digits: μ = 0.643425504760055106... Within-kernel spread: 2.22×10^{-16} (≤ 1 ULP, agreement at the floor of binary64 precision). The Cohn (catalog) holder, recomputed in the same five kernels using a local working copy not redistributed in our companion repository, reproduces byte-exact to μ = 0.643427715576853 with a stable worst pair across all five kernels.

A structural observation about the worst pair. All five verification kernels report the same worst pair (1, 24) for the submission, with the coherence value reproduced to within one unit in the last place of binary64. This is in contrast to the (3, 14) submission in our companion repository, where the five kernels report five different argmax pairs at the same μ value and we interpret that observation as evidence of a degenerate basin floor with a small discrete family of critical points. We interpret the stability of the (1, 24) pair across kernels in this submission as evidence that the basin floor for cell (4, 48) supports a non-degenerate critical structure with a single unambiguous maximum-coherence pair. This is an interpretation consistent with the observations, not a proven structural claim.

4. Apparent basin floor: evidence and limits

Beyond reporting the improved packing, the companion paper Paper_4x48.md in the repository documents a numerical investigation that we interpret as suggesting the submission may lie at a local floor of the basin in which the optimisation was conducted. The lines of evidence are: (i) multi-seed convergence to the same 15th decimal across eight independent initialisations; (ii) stable worst-pair structure across five independent verification kernels (§3); (iii) trust-region radial expansion refutation across 88 attempts at radii spanning five orders of magnitude (1e-6 to 1e-1), with zero commits to a deeper coherence value; (iv) targeted-vector perturbation refutation across single-vector, pair-swap, and hub-targeted attempts, again with zero commits; and (v) 150-fold cold-start enumeration with no seed producing a deeper packing than the submission.

We frame these observations as a working hypothesis consistent with the accumulated evidence, not as a proven theorem. The 8.65% gap to the Levenshtein-2 lower bound 0.5877538100 is mathematically large enough to admit the existence of deeper basins reachable via algebraic seeds, structured constructions, or paradigms not yet considered in this work — including Hoggar-type equiangular tight frame embeddings, mutually unbiased basis embeddings, and cyclotomic-projection constructions. A future result that descended below μ = 0.643425504760055 in this basin would not contradict any fact asserted in this note; it would refine the interpretation we have offered.

5. Repository and reproducibility

The submission file, the companion paper, the headline-numbers document, and the independent Python verifier are publicly available under MIT licence in a dedicated subfolder of the author's repository:

https://github.com/tretoef-estrella/sloane-coherence-records/tree/main/cell_4x48

A reviewer can reproduce the verification in under one minute:

```
git clone https://github.com/tretoef-estrella/sloane-coherence-records.git
cd sloane-coherence-records
python3 verify_sloane_independent.py cell_4x48/4x48_record1.txt
```

Expected output: Coherence mu(Phi) = 0.643425504760055, with the unit-norm check passing within 1×10^{-10} and worst pair (1, 24). The independent Python verifier uses no project-internal code paths; it loads the packing as raw text, reconstructs the Gram matrix using its own routines, and computes the coherence directly.

6. Acknowledgments

We thank Henry Cohn for the published catalog entry for cell (4, 48) in the Game of Sloanes catalog against which this submission is compared; the Cohn holder has been the standing record for this cell for a number of years, and its structural properties (saturated-pair topology, byte-exact unit-norm structure) are the reference baseline throughout this work — Professor Cohn separately performed the independent verification of an unrelated submission for cell (4, 64) in our companion repository's history; Emily King, John Jasper, and Dustin G. Mixon for maintaining the Game of Sloanes reference repository and the published acceptance rule on which this submission's eligibility is based; Henry Cohn, Abhinav Kumar, Gregory Minton for the body of work on coherence packings that informs the methodology, including the linear-programming and semidefinite-programming bound frameworks that constrain where deeper basins could conceivably lie; Welch, Levenstein, Delsarte, Goethals, Seidel, Calderbank, Casazza, Fickus, Tremain, Sloane, and Conway for the foundational results on coherence bounds, Grassmannian packings, and equiangular tight frames on which this submission rests; and Anthropic for building Claude, the AI assistant that collaborated on the structural analysis, the five-kernel verification pipeline, and the basin-floor reasoning.

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